

Implicit parallelism in R

Implicit parallelism is a feature of the R programming language that allows an interpreter to automatically take advantage of the parallelism inherent in some of the R constructions. To support parallel execution, a pure implicitly parallel language does not require any extra directives, operators or functions. R programming achieves implicit parallelism through a number of packages.

- **pnmath:** This implements parallelised versions of the math library's non-RNG routines. For implicit parallelism, it takes advantage of the OpenMP parallel processing directives. When you load the package, the parallel versions of the built-in math functions take their place.
- **Rdsm:** This provides a programming environment for R that is similar to that of threads. The package provides a cleaner, more concise shared memory

world view for R programmers, as well as better performance in some circumstances. It also allows for concurrent processing of very huge, out-of-core matrices.

- **targets:** This examines the dependencies between workflow tasks, skips steps that are already complete, performs computations with optional parallel workers, abstracts files as R objects, and gives physical evidence that the

results match the underlying code and data.

To manage data efficiently, Big Data requires parallel processing technologies. With a small number of packages, R programming allows both implicit and explicit parallelism. We've covered some of the most popular R packages in this article, which can be quite useful for programmers who want to efficiently manage parallelism. **END** 

References

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With pay per use metered costing, clouds are a great option to implement plans without any compromises. The challenges of security and egress costs can be addressed if the cloud is used judiciously to store data. One of the biggest benefits of moving to the cloud is the availability of multiple tools which otherwise have to be bought at very high licensing costs. For storing data, tiering can be easily implemented. The performant data used by applications is only a fraction of the total size of data. Most of this data is unproductive and unemployed but is a resource guzzler. The cold storages like Coldline in Google Cloud or Glaciers in AWS provide immense cost incentives for data that is minimally accessed. This, when coupled with a deduplication tool to eliminate duplicate data and a data processing program to extract relevant

data and discard the fat, is a very simple yet cost-effective strategy to protect data for the long term.

With many public cloud vendors spreading their data centres in all major cities across the globe, the issue of data control is addressed to a large extent. Users can not only ensure their data resides in the required geographical regions but can also have multiple copies of data across considerable distances, making disaster recovery an easy goal to achieve compared to the Herculean task it used to be in the pre-cloud world.

The HBR article talks about single source of truth (SSOT) and

multiple versions of truth (MVOT). SSOT data has to be incorruptible and immutable in some cases, with a very high degree of provenance and governance leading to its indisputable integrity. A lack of SSOT leads to a dent in an organisation's credibility. Clouds play a major role in not only building a SSOT repository but also in providing a plethora of choices. These include Google Data Studio and Bigquery or Microsoft's PowerBuilder, which help to build an offensive strategy, which can then foster the development of effective and insightful platforms for MVOTs, thus propelling the organisation to the next level of growth. **END** 

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